



Data structure and Programming II

TP11: Sorting2

- Objective:
 - To practice sort data using various sorting algorithms
 - Merge Sort, Quick Sort
 - Apply C++ programming
- Individual work
- Submit to Moodle (do not zip)
 - *A pdf report (cover, exercise and solution with screenshot)*
 - *Source codes*

Deadline:
6 days

1. Write a C++ program to sort 5 integer numbers stored in an array from user input using Merge sort algorithm. Display result on screen.

```
Input number #1: 4
Input number #2: -2
Input number #3: 0
Input number #4: 7
Input number #5: 20

Merge Sort : -2 0 4 7 20
```

2. Read data from the txt file below and store in an array. Sort that array using the Merge sort algorithms and write sorted array into new files namely selection.txt.

```
99
2
7
8
0
-1
99
1
5
```

3. Write a C++ program to sort 5 integer numbers stored in an array from user input using Quick sort algorithm. Sort from largest to smallest. Display result on screen.

```
Input number #1: 4
Input number #2: -2
Input number #3: 0
Input number #4: 7
Input number #5: 20

Quick Sort: -2 0 4 7 20
```

4. Given a student list below. Read data from csv file into an array. Sort the data in array using Quick algorithm.

The program should be able to sort the data by ... (provide a menu of choice to a user)

- a. Sort by ID
- b. Sort by name
- c. Sort by gender

ID	Name	Gender	DoB
1010	Sok	M	12-Jan-93
1009	Sokha	F	11-Feb-90
1025	Dara	M	10-Jan-91
1023	Sitha	F	21-Sep-90
1000	Jack	M	22-Sep-81
1002	Rose	F	23-Oct-90
1003	Ratha	M	20-Dec-91